

Lecture 21 Normal Approximation To Binomial Distribution & Sampling Distribution of The Sample Proportion

BIO210 Biostatistics

Xi Chen

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School of Life Sciences

Southern University of Science and Technology

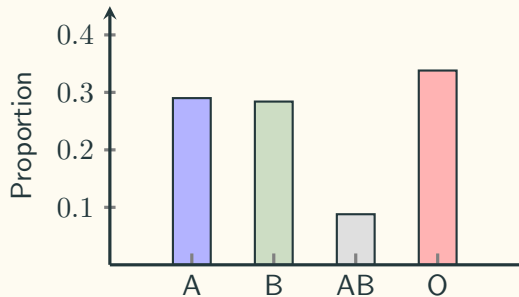


南方科技大学生命科学学院
SUSTech · SCHOOL OF
LIFE SCIENCES

ABO Blood Types Proportions In Han Chinese

Population distribution of ABO blood types in Han Chinese.

Total	A	B	AB	O
592,243	171,473	168,040	52,088	200,642
1	0.290	0.284	0.088	0.338



中国汉族人ABO血型的分布

空军成都医院* 彭德仁

研究 ABO 血型的分布在医学、法医学及人类学等方面都有重要意义。关于国人的 ABO 血型分布早在 1918 年就有人报道过^[1]。现已积累了大量的资料。1963 年，尚书颂等曾统计分析了 15 万多中国人 ABO 血型的分布资料，提出将我国各省区的 ABO 血型分布划分为 4 种类型^[2]。1982 年，陈雅勇等收集了 1920~1979 年国内外发表的中国人的 ABO 血型分布资料共 28 万多例，通过计算各群体间的遗传距离，将全国分为 4 个组^[3]。但这两篇文献都包含了少数民族的资料。Mourant 等所著的《人类血型分布》中也仅收集到 18 万多中国人的 ABO 血型分布资料^[4]。由于中国是一个多民族的国家，就 ABO 血型而言，不同的民族可有不同的分布特点，即使是同一民族，其分布特点因地域等原因也可能不尽相同。为了给医学、法医学及人类学等研究提供一些基本数据，本文收集了 1920~1988 年国内外发表的有关汉族的 ABO 血型分布资料共 59 万多人，并对其进行统计分析。

材料与方法

(一) 资料来源

国内发表的资料主要取自 1963~1966 年的《天津医药杂志输血及血液学附刊》、1978~

1979 年的《输血及血液学》杂志、1980~1988 年的《中华血液学杂志》、1981~1988 年的《中华医学检验杂志》等的 162 篇文献；国外发表的资料主要取自《人类血型分布》^[4]。所收集的资料仅限于汉族，每份资料的人数均多于 30 人且注明了居住地区，全部资料共计 1 022 237 人。

(二) 基因频率的计算与 Hardy-Weinberg 吻合度测验

对所有收集到的有关资料用赵桐茂推荐的方法来计算 ABO 基因频率^[5]。p、q、r 分别代表 A、B、O 基因频率。为了估计调查资料的可靠性，对每份原始资料均作了显著性测验^[6]。如果 $|D/b| \leq 2$ ，表示观察值与期望值无显著性差异，此时 $P \geq 0.05$ （即 Hardy-Weinberg 吻合度测验观察值与期望值吻合度很好）；如果 $|D/b| > 2$ ，则 $P < 0.05$ ，表示观察值与期望值有显著性差异。AB 型的观察值小于期望值，D/b 为正值，反之则为负值。所收集到的资料中除去 $|D/b| > 2$ 的 81 份外，最后所选择的 327 份的数据按地区合并，并根据 Hirschfeld 等提出的 $(A + AB)/(B + AB)$ 公式计算民族指数。但该指数只反映 A 和 B 基因的比例，并不能反映差异程度^[7]。

(三) 遗传距离

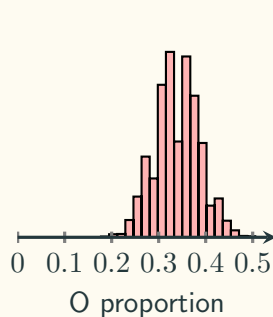
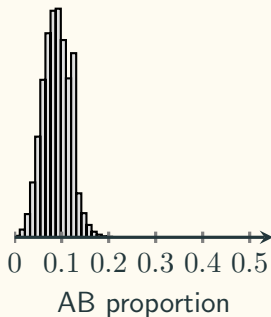
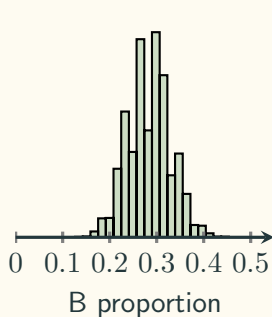
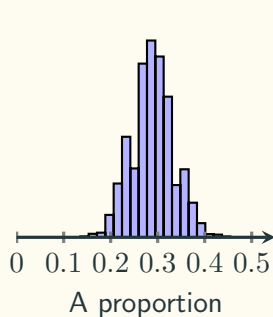
为比较 ABO 血型分布在各地区间的差异，使用遗传距离 d，其公式为 $d = 4(1 - \cos \theta)/\pi$

* 邮政编码 610061

Sampling Distribution of ABO Blood Type Proportions

Population distribution of ABO blood types in Han Chinese.

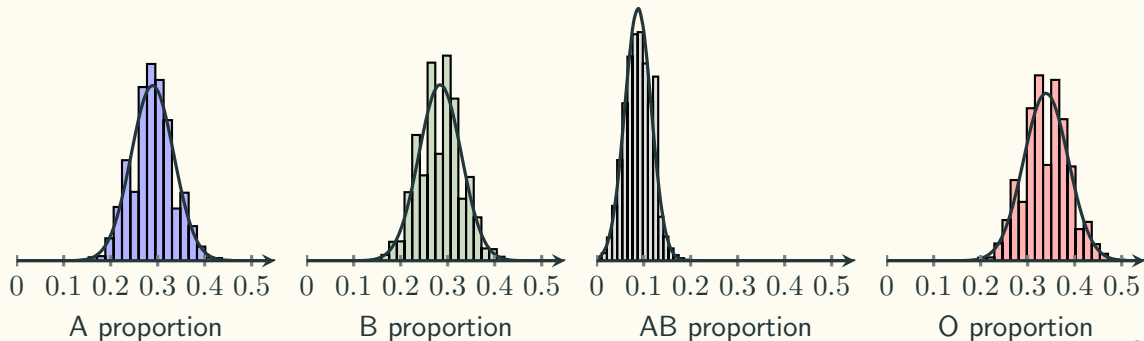
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Sampling Distribution of The Sample Proportion

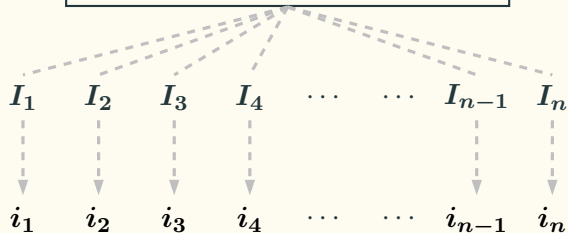
Fraction of type A blood in the population: π

A, A, B, O, O, AB, O, A, O, B, ...

Sampling Distribution of The Sample Proportion

Fraction of type A blood in the population: π

A, A, B, O, O, AB, O, A, O, B, ...



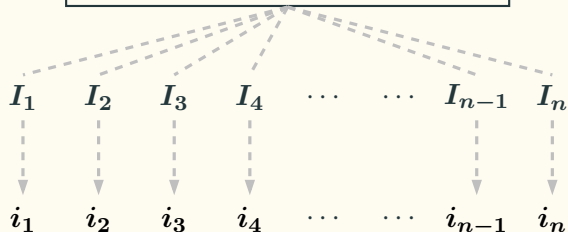
$$Y = \sum_{i=1}^n I_i \quad \bar{I} = \frac{1}{n}Y$$

Dashed arrows point from the summation symbol in the first equation to y , and from the $\bar{I}$ term in the second equation to $\bar{i}$.

Sampling Distribution of The Sample Proportion

Fraction of type A blood in the population: π

A, A, B, O, O, AB, O, A, O, B, ...



$$Y = \sum_{i=1}^n I_i \quad \bar{I} = \frac{1}{n}Y$$

Dashed arrows point from Y to y and from $\bar{I}$ to $\bar{i}$.

sample

The Sum And The Mean of Indicator Variables

I: Indicator Variable

i.i.d.

$$I_1 \sim \text{Ber}(\pi)$$

$$I_2 \sim \text{Ber}(\pi)$$

$$I_3 \sim \text{Ber}(\pi)$$

$\vdots$

$\vdots$

$$I_{n-1} \sim \text{Ber}(\pi)$$

$$I_n \sim \text{Ber}(\pi)$$

The Sum And The Mean of Indicator Variables

Meaning of Y : number of people with blood type A per n people.

Meaning of $\bar{I}$: The proportion of people with blood type A.

I : Indicator Variable

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$$Y = \sum_{i=1}^n I_i \sim ? \quad \bar{I} = \frac{1}{n} Y \sim ?$$

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By definition:

$$Y \sim B(n, \pi)$$

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By definition:

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By The Central Limit Theorem:

$$\bar{I} \dot{\sim} \mathcal{N} \left(\mu = \pi, \sigma^2 = \frac{\pi(1-\pi)}{n} \right)$$

$$Y = n\bar{I} \dot{\sim} \mathcal{N} (\mu = n\pi, \sigma^2 = n\pi(1-\pi))$$

Normal Approximation To A Binomial Distribution

Our knowledge about Han Chinese (Peng, 1991):

Total	A	B	AB	O
1	0.290	0.284	0.088	0.338

A sample from Wuhan (Xu et al., 2015): 1,188 out of 3,694 people have blood type A.

Questions:

1. When draw a random sample ($n = 3,694$), what is the probability of getting 1,100 – 1,200 people with blood type A?
2. When draw a random sample ($n = 3,694$), what is the probability of getting 1,188 people with blood type A?

Normal Approximation To A Binomial Distribution

Question 1:

Use the Binomial probability :

$$\sum_{k=1100}^{1200} \binom{3694}{k} 0.29^k 0.71^{3694-k} = 0.152949$$

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Use the Normal probability :

$$\mathbb{P}(1100 \leq x \leq 1200) = \mathbb{P}\left(\frac{1100 - 1071.26}{27.58} \leq z \leq \frac{1200 - 1071.26}{27.58}\right) = 0.148681$$

Normal Approximation To A Binomial Distribution

Question 1:

Use the Binomial probability :

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Use the Normal probability :

$$\mathbb{P}(1100 \leq x \leq 1200) = \mathbb{P}\left(\frac{1100 - 1071.26}{27.58} \leq z \leq \frac{1200 - 1071.26}{27.58}\right) = 0.148681$$

Use the Normal probability with **continuity correction** :

$$\mathbb{P}(1100 - 0.5 \leq x \leq 1200 + 0.5) = 0.152923$$

Normal Approximation To A Binomial Distribution

Question 2:

Use the Binomial probability :

$$\binom{3694}{1188} 0.29^{1188} 0.71^{3694-1188} = 2.16 \times 10^{-6}$$

Use the Normal probability with continuity correction :

$$\mathbb{P}(1188 - 0.5 \leq x \leq 1188 + 0.5) = 1.86 \times 10^{-6}$$

Sampling Distribution of The Sample Proportion

- $\bar{I} \sim$ **Sampling Distribution of The Sample Proportion**
- Generally, we used $p = \frac{x}{n}$ to represent the sample proportion, which is an point estimate for the population parameter π .
- According to the **Central Limit Theorem**, when the sample size n is large enough, we have:

$$\mathbf{P} \dot{\sim} \mathcal{N}(\mu_{\mathbf{P}}, \sigma_{\mathbf{P}}^2), \text{ where } \mu_{\mathbf{P}} = \pi, \sigma_{\mathbf{P}}^2 = \frac{\pi(1 - \pi)}{n}$$

Sampling Distribution of The Sample Proportion

